Measurement of Three-Dimensional Unsteady Flows Using an Inexpensive Multiple Disk Probe

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ABSTRACT

A novel velocimeter consisting of multiple orthogonal disks fitted with pressure transducers has been developed. In such a velocimeter the pressure difference is measured between the center of each disk face and the center of its other face for each of the three orthogonal disks. The three components of fluid velocity can be deduced from the three measured pressure differences. While previously developed anemometers based on dynamic pressure differences (such as yawhead or five-hole probes) can only measure velocities with a small range of directions, the new disk probe can measure three components of velocity, even in highly three-dimensional flows where the approximate direction of the flow is not known. Previous work demonstrated that in steady flows the device could measure velocities to $\pm 1.4\%$ and angles to $\pm 4°$. In the present work, involving both field trials and wind tunnel tests, it is shown that the disk probe can measure three-dimensional unsteady flows with accuracy suitable for many meteorological applications. The disk probe tested has a flat frequency response up to 3 or 4 Hz and can measure velocity magnitudes with an accuracy of better than ± 0.3 m s^{-1}. Simple modifications to the disk probe would increase its frequency range to 10 Hz or better.

1. Introduction

Measurement of atmospheric and oceanic flow fields is challenging because conditions are often harsh, and the flows are often highly three-dimensional with a broad range of spatial and temporal scales. The energy-containing eddies have a timescale of more than 1 s, while the dissipation occurs on timescales of 10–100 Hz (Lutgens 1998; Stull and Ahrens 1995).

Depending on the measurement requirements, meteorological flows can be measured using a number of physical principles (De Felice 1998). Accurate, fast measurements can be made with devices such as hot wire anemometers (Comte-Bellot 1976; Perry 1982), laser Doppler velocimeters (Drain 1980; Durst et al. 1981), particle image velocimeters (Adrian 1991; Grant 1997), and a host of others. These devices are characterized by excellent accuracy (typically, velocities can be measured to within 1%) but are costly ($10 000–$100 000 U.S.), complex, and have comparatively poor durability. Their high cost and poor durability in general limit these devices to use in a controlled environment like a laboratory; they are rarely used in an industrial or a field work setting. Currently, sonic anemometers are used for meteorological studies in which good time resolution is required, but the cost of these instruments (≈$20 000 U.S.) is also high and precludes their use in many applications. Airfoil probes, thermistors, and cold film probes have been used to measure turbulence spectra in water, but they cannot measure directional velocity fluctuations and are not suitable for meteorological applications.

Devices that measure dynamic pressure (or force) can be made less expensively. Such instruments include cup, propeller, and vane anemometers (Wyngaard 1981); claw probes; yaw head probes; and five-hole and seven-hole probes (Rae and Pope 1984; Chue 1975; Zilliac 1993; Everett et al. 1983). In general, these devices are reliable and reasonably accurate (directions to $\pm 1°$ and velocities to $\pm 1\%$) but can accurately measure only mean flow velocities and must be aligned to within $\pm 70°$ of the flow direction (Chue 1975; Zilliac 1993). Fritschen and Gay (1979) present an overview of these classic mechanical devices. Recently, Rediniotis and Kinser (1998) have developed a probe using 18 holes over the surface of a sphere. Although this probe is nearly omnidirectional (cone angles $>90°$) and is apparently accurate for the reported conditions, calibration of that device is complex due to the large number of pressure taps and the varying flow regimes that occur around a sphere at different Reynolds numbers.

In a previous study, Green and Rogak (1999) showed that the pressure differentials across three orthogonal disks in a flow could be used to determine three velocity

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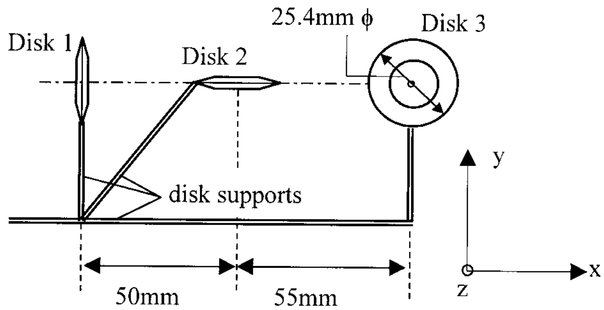

FIG. 1. Geometry of the multiple disk probe. Disk 3 and the normal to disk 1 lie in the same (x–y) plane. Disk 2 is 90 mm above that plane. All disks are 3.5 mm in thickness.

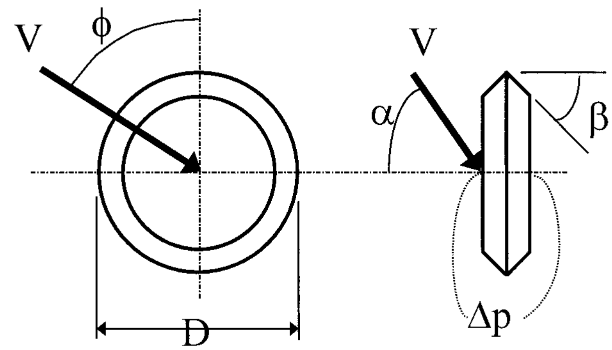

FIG. 2. The geometry of a single disk (a) face on and (b) side view. Due to symmetry, Δp is not a function of the azimuthal angle ϕ. The disk bevel angle is β.

components. Such a device was shown to be very accurate for measurements of two-dimensional steady flow fields in a laboratory. These encouraging results suggested that the disk probe might be developed for three-dimensional atmospheric measurements, assuming that the device could be made suitably robust for field work and that the frequency response was suitable.

The disk probe is inherently robust because it has no exposed moving parts; however, it is possible that the pressure taps could become blocked by dirt or spray. This is a problem facing other devices that measure pressure (e.g., Pitot probe for dynamic pressure and the Elliot probe for static pressure), many of which are routinely used in field conditions. For extreme conditions, methods have been developed (Donelan et al. 1999) to automatically remove droplets and dirt from the pressure taps in field conditions.

The frequency response of the disk probe is difficult to predict and somewhat complicated to measure under field conditions because 1) existing ''standard'' instruments have their own limitations and 2) instruments placed side by side will experience slightly different velocity fields due to the physical separation. In the present work, the multiple disk probe performance is compared to the performance of a sonic anemometer.

2. Theory of operation of a multiple disk probe

In a multiple disk probe, three orthogonal disks are arranged as shown in Fig. 1. Each disk (Fig. 2) is fitted with a flush-mounted pressure transducer, or pressure tap lines connected to a pressure transducer, to measure the pressure difference from the center of one side of the disk to the center of the opposite side. Neglecting the (small) effect of fixtures required to hold the disks in place, the pressure difference between one side of the disk and the other side must be a function solely of the disk geometry, the properties of the air, the magnitude of the velocity, and the angle made by the velocity vector relative to the normal vector to the disk. Most significantly, owing to the symmetry of the disk geometry, the pressure difference cannot be a function of the azimuthal orientation of the velocity vector $\boldsymbol{\phi}$.

Measurement of velocity fields with the multiple disk probe requires accurate characterization of the behavior of a single disk. The aerodynamic characteristics of individual disks were investigated by Green and Rogak (1999). Additional measurements, conducted on a 30% scale model of a single disk, show that the pressure coefficient is nearly constant for Re $>$ 4000 (Fig. 3), which simplifies the data analysis described below.

The principle of operation of a multiple disk probe is best understood by considering a purely two-dimensional flow (Fig. 4). Qualitatively, if $0 < \alpha_1 < \pi/2$, then $(\Delta p)_1$ is positive and so is $(\Delta p)_2$. For $\pi/2 < \alpha_1 < \pi$, $(\Delta p)_1$ is negative and $(\Delta p)_2$ is positive. Similarly, for $\pi < \alpha_1\ \pi/2$, $(\Delta p)_1$ is negative and so is $(\Delta p)_2$. Finally, for $3\pi/2 < \alpha_1 < 2\pi$, $(\Delta p)_1$ is positive and $(\Delta p)_2$ is negative. Therefore, it is trivial to distinguish the quadrant of the flow direction. Determining the exact angle of the flow is slightly more involved.

In view of the physical argument made above, the pressure difference across disk 1, $(\Delta p)_1$, is, at low Mach number, a function of the disk geometry (which can be characterized by the disk diameter, D); the magnitude of the flow velocity, V; the density of the fluid, ρ; the dynamic viscosity of the fluid, μ; and the angle made by the flow to the disk 1 normal, α_1. In nondimensional form, this information may be expressed as $(c_p)_1 = f(\mathrm{Re}_D, \alpha_1)$ where f is some function, $c_p = \Delta p/(0.5\rho V^2)$ is the pressure coefficient, and $\mathrm{Re}_D = \rho VD/\mu$ is the Reynolds number based on disk diameter. Similarly, for the second disk, $(c_p)_2 = f(\mathrm{Re}_D, \alpha_2)$. Now since $\alpha_2 = (\pi/2) - \alpha_1$, if one measures the two pressure coefficients, one effectively has two equations to solve for the two unknowns: Re_D and α_1. Alternatively, one may consider a flow with some fixed viscosity and density about two disks of the same diameter. Then, $(\Delta p)_1 = g(V, \alpha_1)$ and $(\Delta p)_2 = g[V, (\pi/2) - \alpha_1]$, where g is some function, and again it is clear that by measuring $(\Delta p)_1$ and $(\Delta p)_2$, one has two equations in the two primitive variables V and α_1.

As will be discussed below, owing to the one-to-one dependence of (c_p) on α_1 and Re_D separately, the two

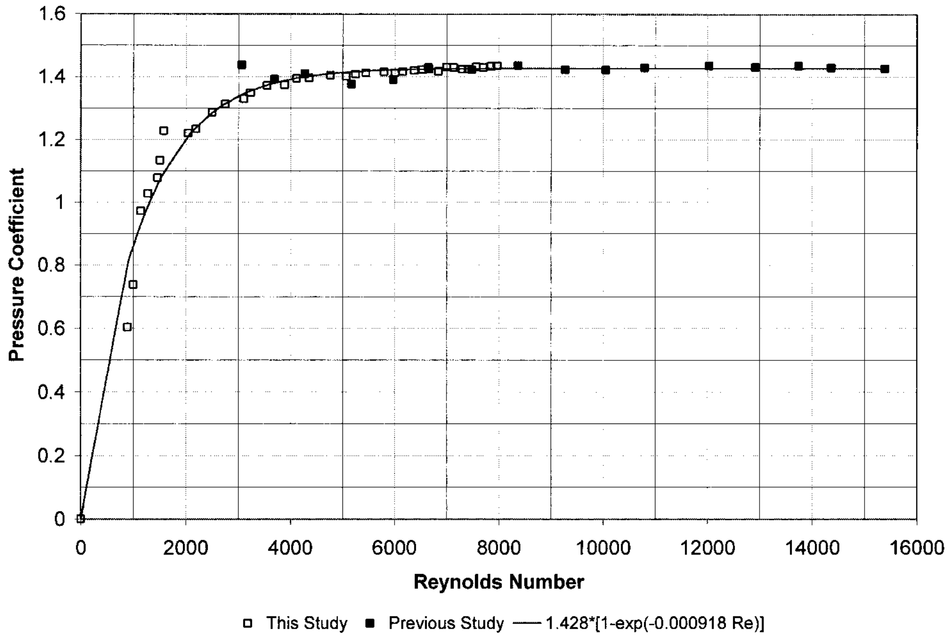

FIG. 3. The dependence of the disk pressure coefficient, $c_p = \Delta p/(0.5\rho V^2)$, on the Reynolds number. The disk is normal to the flow direction.

equations described above may be solved readily and yield a unique solution. In particular,

$$(\Delta p)_1 = \frac{1}{2}\rho V^2 f(\mathrm{Re}_D, \alpha_1), \tag{1}$$

$$(\Delta p)_2 = \frac{1}{2}\rho V^2 f\left(\mathrm{Re}_D, \frac{\pi}{2} - \alpha_1\right). \tag{2}$$

Dividing Eq. (1) by Eq. (2) yields a function h that depends on α_1 and Re_D:

$$\frac{(\Delta p)_1}{(\Delta p)_2} = \frac{f(\mathrm{Re}_D, \alpha_1)}{f[\mathrm{Re}_D, (\pi/2) - \alpha_1]} = h(\mathrm{Re}_D, \alpha_1). \tag{3}$$

It turns over that h depends weakly on Re_D and varies almost monotonically with α [see Fig. 5 of Green and Rogak (1999)]. Using any reasonable initial guess for Re_D, it is possible to estimate the flow angle by measuring h. Equation (1) or (2) can then be used to determine the airspeed V, which can be used to calculate Re_D. Flow angle and speed are then refined. This procedure converges in one iteration. Extension of this method to three-dimensional flows measured by three orthogonal disks is fairly straightforward because the pressure coefficient for each disk exhibits a ''cosine-like'' variation with α. Initially, the individual velocity components are estimated by assuming unity pressure coefficients for each disk. These estimates are used to calculate the Reynolds number and flow angles to each disk. A new pressure coefficient is calculated for each disk using the single-disk calibration data, resulting in new velocities and angles. Several iterations are required, but the data processing can proceed in real time on a modern personal computer (refer to the appendix). Alternatively, a simple look-up table may be used to yield accurate velocity results.

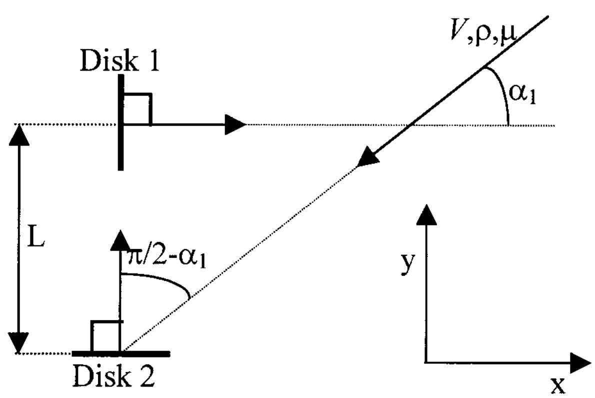

FIG. 4. Determination of two-dimensional wind velocity using two disks.

3. Experimental work

a. Description of the multiple disk probe

The particular multiple disk probe used in the tests described below consisted of three orthogonal disks spaced to fit within a rectangular prism 12 cm × 10.5 cm × 3 cm. Each disk was 25.4 mm in diameter, 3.5 mm in thickness, and had 55° bevelled edges. The pressures at the centers of opposing sides of each disk are sensed by a differential pressure transducer (Omega PX 665). The 4–20-mA output of the transducers was con-

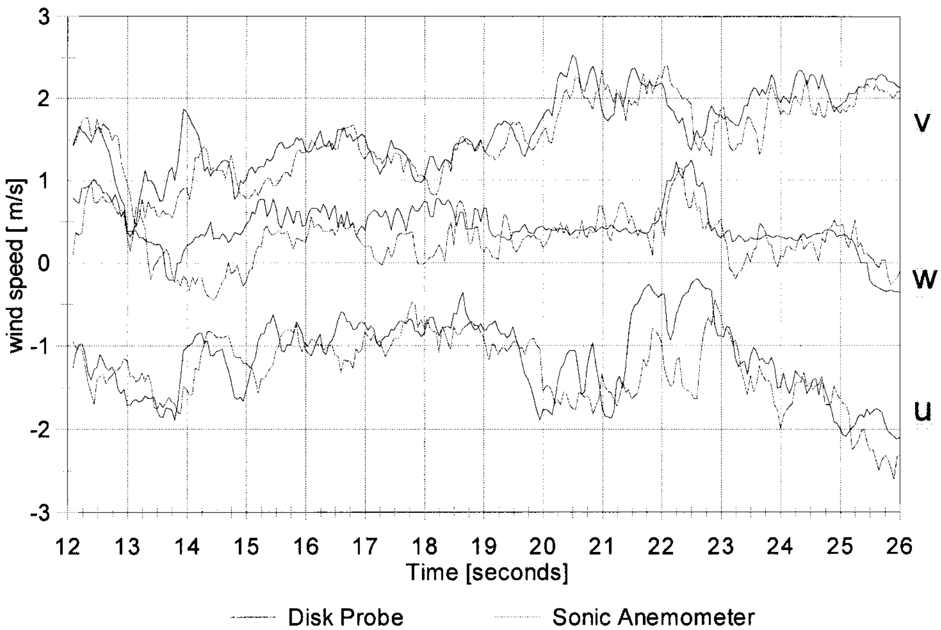

FIG. 5. Comparison of instantaneous velocity components measured by the disk probe anemometer and the sonic anemometer during field trials.

verted to a voltage by means of a series precision 250Ω resistor. Stainless steel syringes mounted flush with the disk faces (the disks were milled out and then back filled with epoxy around the syringes to accomplish this), linked to 2-mm-diameter plastic tubing, provided the connection between the pressure taps at the center of the disk faces and the pressure transducers.

The frequency response of the disk connected to the pressure transducer was measured by placing the disk probe in the wake of a large object in the wind tunnel, then suddenly removing the obstacle. These tests showed that the disk–transducer combination has a time constant of 106 ms, which suggests that it cannot reliably measure fluctuations higher than 4 Hz. The frequency response of the disk–transducer combination is primarily a function of the transducer air volume and the diameter of the capillary tube connecting the transducer to the probe and not a fundamental characteristic of the disk itself. Although the frequency response of the probe can be improved, it is already better than that of a lightweight propeller anemometer.

b. Field tests

Field tests were conducted in an open area at the south end of the University of British Columbia (UBC) campus. The disk probe and a 3-axis ultrasonic anemometer (3-axis Research Acoustic Anemometer, produced by Gill Instruments) were mounted on a meteorological tower 2.5 m off the ground with a horizontal distance of 0.5 m between the instruments. To avoid interference from the tower wake, both instruments were mounted on a 2-in. pipe at least 1 m from the tower, with the pipe oriented approximately perpendicular to the mean wind direction. Data were recorded with a Campbell Scientific 21X Micrologger connected to the RS232 port on a laptop PC.

c. Wind tunnel tests

The wind velocities during the field trials were relatively low, as unfortunately UBC experiences a large number of calm hours, and neither the disk probe nor the sonic anemometer was available for long-term monitoring studies. In March 2000, a wind tunnel study was conducted to compare the two instruments in an unsteady flow with higher velocities. These studies were carried out in the UBC Boundary Layer Wind Tunnel. The unsteady flow field was created in two ways: by varying the wind tunnel speed manually and by placing large objects in the tunnel to generate an unsteady wake. The anemometers were placed symmetrically in the wake downstream of these objects, either in the tunnel or downstream of the tunnel exit.

4. Results and discussion

a. Field trial and wind tunnel results

Figure 5 compares the u, v, and w wind components measured by the sonic anemometer and the disk probe during one of the field trials. Agreement is very good for all three velocity components for the great majority of the test period. Other field trials showed comparably good agreement between the two devices. Perfect agreement would not be expected because of the spatial sep-

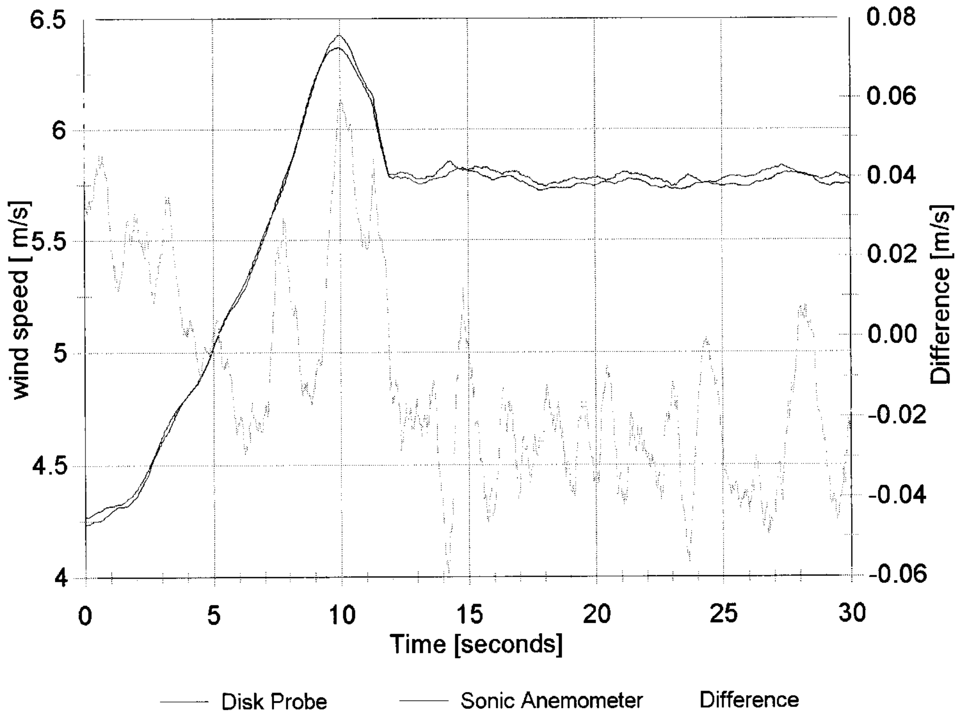

FIG. 6. Comparison of instantaneous streamwise velocity component measured by the disk probe anemometer and the sonic anemometer during wind tunnel tests.

aration between the two instruments. For this reason, it is perhaps more useful to compare wind velocity statistics (mean flow direction, mean flow speed, and the frequency spectrum) that should be identical at the two instrument locations, and this is done in section 4b below.

Figure 6 compares typical measurements of one velocity component during one of the variable wind speed tests. The relative turbulence intensity in the wind tunnel tests was generally less than in the field studies, but the fluctuations at the two anemometers show excellent agreement. The time-averaged difference in velocity between the disk probe and sonic anemometers is a remarkably low 0.01 m s^{-1}, and the peak wind speed difference is only 0.06 m s^{-1}. All the other wind tunnel trials show similarly good agreement.[1]

b. Comparison of flow field statistics

Due to the physical and/or temporal separation of the anemometers during the field and wind tunnel tests, the true flow fields were not identical for the two instruments. For meteorological studies, one would desire that the different anemometers provide the same wind velocity, direction, and spectra. Figure 7 compares the average wind speeds from the 4 field trials and 12 wind tunnel tests. Even including the field trial 2 results, the agreement is nearly perfect (the best-fit line has a slope of 0.994 with a standard error in the slope of 0.006; the y intercept is 0.05 m s^{-1} with a standard error of 0.09 m s^{-1}, and the correlation coefficient is 0.9993), within the accuracy of the ‘‘gold standard’’ sonic anemometer. It should be emphasized here that the disk probe measurements were calculated from wind tunnel calibrations of single disks, and no adjustment was performed to improve the agreement with the sonic anemometer.

Comparison of the average horizontal wind angle (Fig. 8) also shows good agreement (the best-fit line has a slope of 1.05; the y intercept is $-1°$; and the correlation coefficient is 0.989), considering that the physical alignment of the two anemometers might have had an error of up to 4°. The large standard deviations of the wind direction readings are evidence of the highly unsteady character of these flows. The nearly equal lengths of the vertical and horizontal ‘‘error bars’’ imply that the disk probe and sonic anemometers measured comparable amounts of flow unsteadiness.

The frequency response of the two devices was analyzed using the fast Fourier transform (FFT). Figure 9 compares the frequency response of the disk probe and the sonic anemometer between 0 and 10 Hz (the 10-Hz

[1] For a period during the wind tunnel tests, one component of the disk probe–measured velocity was too high relative to the sonic anemometer by a constant factor of 1.18; suddenly during one test, the factor returned to 1.00 and remained there for all subsequent tests. We believe that this was due to an electrical ground malfunction and corrected the channel by 1.18 in the plots shown here. The corrected results (labeled as WT2-7 in Figs. 7 and 8) are consistent with the data for which there was no electrical fault, which implies that this procedure for correcting the data was not unphysical.

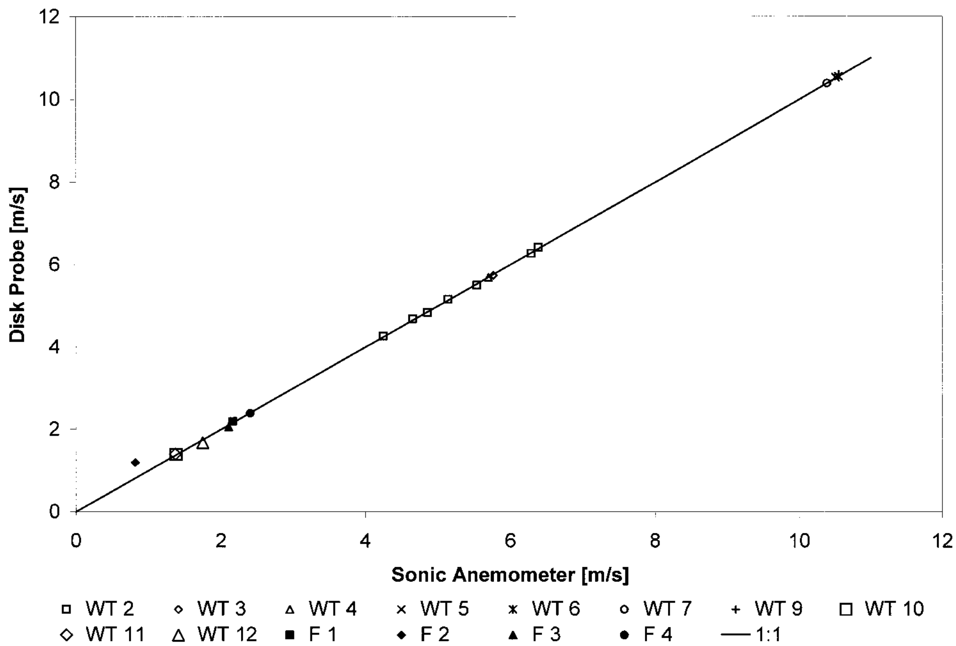

FIG. 7. Comparison of the mean wind speed, measured by the disk probe anemometer and the sonic anemometer, during 10 wind tunnel tests and 4 field trials. The straight line shown represents a perfect 1:1 agreement between the two devices. The agreement between the two instruments is obviously nearly perfect for all trials except for field trial 2 (labeled as F2), during which an electrical fault occurred in the instrumentation.

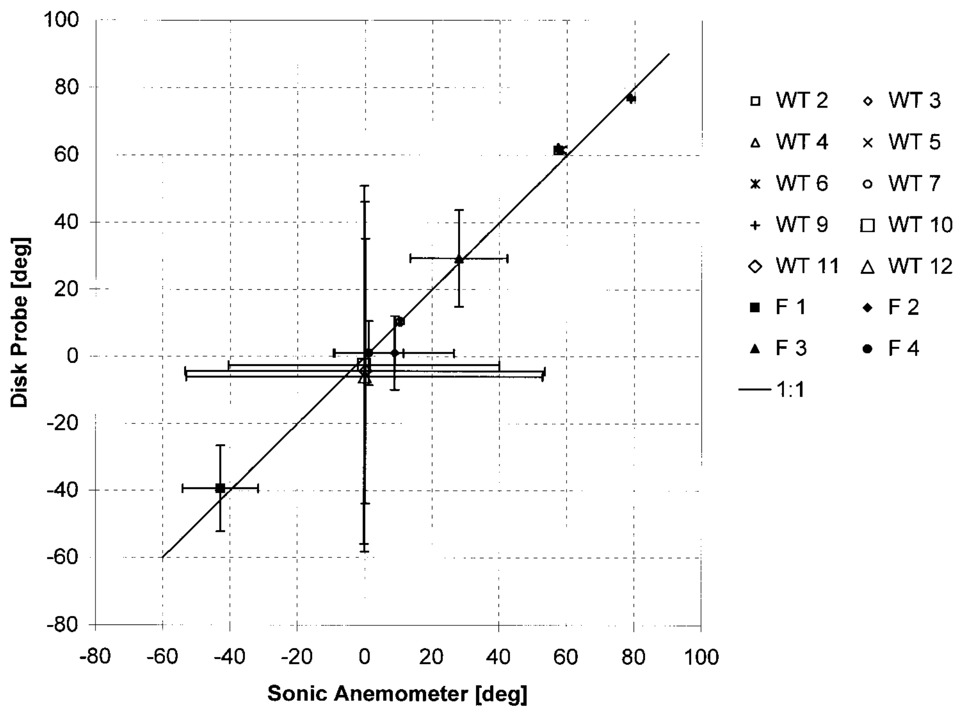

FIG. 8. Comparison of the mean wind direction measured by the disk probe anemometer and the sonic anemometer, during 10 wind tunnel tests and 4 field trials. The error bars represent the std dev in the measured angles (caused by the flow unsteadiness). The straight line shown represents a perfect 1:1 agreement between the two devices.

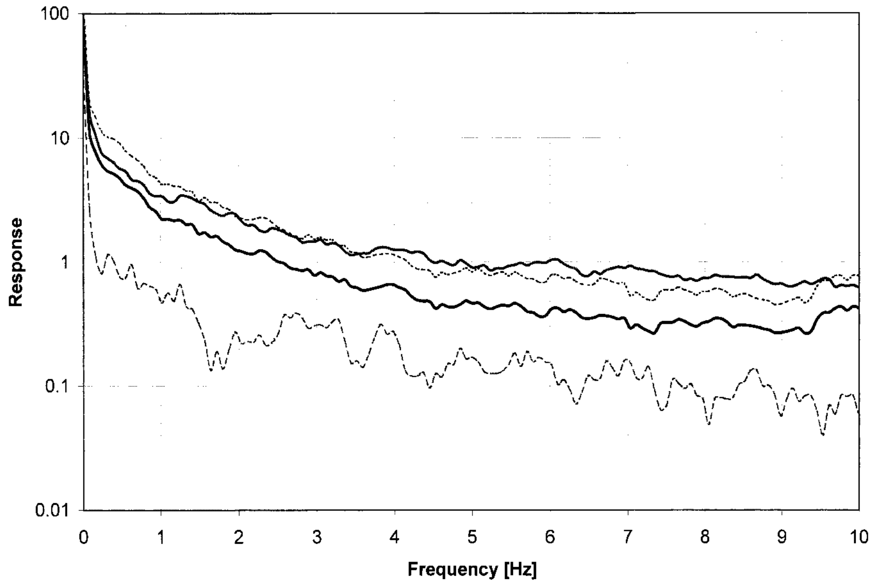

FIG. 9. Ensemble average frequency spectrum of the disk probe and the sonic anemometer during a typical field trial. Dashed curves have been drawn 1 std dev above the disk probe spectrum and 1 std dev below the sonic anemometer spectrum. For all frequencies below 3.5 Hz, the sonic anemometer spectrum is within 1 std dev of the disk probe spectrum, and for all frequencies, the disk probe spectrum is within 1 std dev of the sonic anemometer spectrum.

window chosen was a quarter of the data acquisition frequency). The spectra are in close agreement up to 3 or 4 Hz, which is in line with the independent measurement of probe frequency response described in section 3a.

There are no simple statistical tests for comparing frequency spectra. To get some sense of whether the spectra differences are significant, we performed a number of different tests under uniform conditions with both anemometers and computed the ensemble-average standard deviation of these signals. A curve 1 standard deviation away from both the disk probe and sonic anemometer spectra is shown in Fig. 9. These curves demonstrate that the differences between the sonic anemometer and disk probe spectra are limited to within 1 standard deviation of the spectra. Figure 10 shows a similar result for one of the wind tunnel trials. All spectra were quite similar up to 4 Hz.

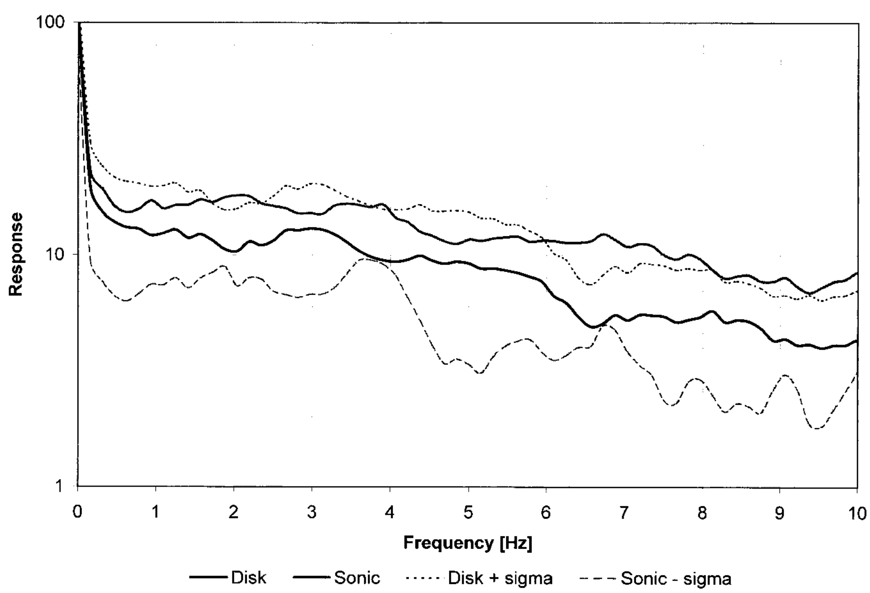

FIG. 10. Frequency spectrum of the disk probe and the sonic anemometer during a typical wind tunnel test. As in the field trial, the spectra are identical to within 1 std dev.

5. Summary

Three-dimensional unsteady flow fields may be measured using a multiple disk probe consisting essentially of three orthogonal disks and three pressure transducers to measure the pressure difference across the three disks. For many field applications, this disk probe anemometer would provide performance equivalent to a sonic anemometer but at a much lower cost. For example, the current design would be well suited to implementation of a large network to provide three component measurements on a second-by-second basis with a spatial density much higher than would be feasible with more expensive instruments.

The present prototype had diminished frequency response above 4 Hz due to limitations of the pressure transducer and the connections to the disk pressure taps. This limitation might be overcome to the point where the response is constrained only by the physical size of the measurement volume, as it is for the sonic anemometer. These improvements, in addition to implementation of existing automatic cleaning systems, would expand the number of applications for the multiple disk probe.

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APPENDIX

Algorithm for Converting Measured Pressures to Flow Velocities

Step 1: Using the pressure readings in the x, y, and z directions (see Table A1), compute the corresponding velocity for each direction with an assumed pressure coefficient of 1. These are the initial velocity estimates. Note that the pressure is treated as strictly positive.

Step 2: If the change in total velocity from the last estimate is small, of a very large number of iterations that have been performed, go to step 8. Otherwise, proceed to step 3.

Step 3: Based on the most current velocity estimates, compute the angle of flow to the normal defining each probe face.

Step 4: Using tabulated data of c_p versus angle for high Reynolds numbers, interpolate between angles to find a revised estimate of c_p for each probe. Use 80% of this new estimate and 20% of the old value to assign to the new c_p. This ensures smooth, stable convergence of the algorithm.

Step 5: Compute the Reynolds number for each probe and refine the c_p estimate using the known c_p versus Reynolds number curve.

Step 6: Compute the new velocity estimates based on the revised coefficients of pressure.

Step 7: Compute the change in the velocity magnitude from the previous iteration. Return to step 2.

Step 8: Check to see if convergence has been reached. If so (convergence was always reached in the experiments we conducted), compute the actual wind direction based on the angle estimates above and the signs of the pressure readings.

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TABLE A1. Convergence of the algorithm for the specific case where the pressure difference across the disk normal to the x direction is 2.49 Pa, the pressure difference across the disk normal to the y direction is 6.21 Pa, and the pressure difference across the "z" disk is 0.249 Pa. The air kinematic viscosity is 1.5×10^{-5} m^2 s^{-1} and the air density is 1.20 kg m^{-3}. The disk diameter is 2.54 cm.

Itera-	Coefficient of pressure			Velocity (m s^{-1})			Flow angle (deg)			Reynolds number		
tion	x	y	z	x	y	z	x	y	z	x	y	z
1	1.00	1.000	1.000	2.035	3.218	0.644	58.2	33.6	80.4	3443	5444	1089
2	0.843	1.220	0.309	1.166	2.424	0.193	64.4	26.0	85.9	1976	4108	327
3	0.611	1.269	0.026	1.124	2.563	0.284	66.5	24.3	84.2	1905	4345	481
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
10	0.518	1.295	0.052	1.131	2.567	0.316	66.4	24.6	83.6	1917	4350	536
11	0.518	1.295	0.052	1.131	2.567	0.316	66.4	24.6	83.6	1917	4350	536

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